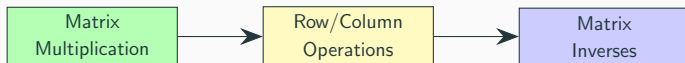


Row and Column Operations

Contravariant vs Covariant Logic

Learning Objectives



Key Questions:

- What are **row operations** and **column operations**?
- Why do rows have **contravariant** logic and columns have **covariant** logic?
- How do **simultaneous operations** preserve equations $AB = C$?
- How to use row/column operations to find **matrix inverses**?

Review: Three Views of Matrix Multiplication

For $C = AB$, we have three interpretations:

1. **Column view:** Each column of C is a linear combination of columns of A

$$|c_j| = A|b_j|$$

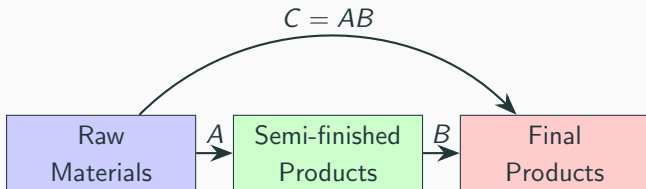
2. **Row view:** Each row of C is a linear combination of rows of B

$$-c_i- = -a_i- B$$

3. **Sum-of-rank-one:**

$$C = AB = \sum_{k=1}^n |a_k| \cdot -b_k-$$

Review: The Ingredient Table Interpretation



Matrix $C = AB$ represents the **composition** of two recipe tables:

- A : Semi-finished $\leftarrow$ Raw materials
- B : Final products $\leftarrow$ Semi-finished
- $C = AB$: Final products $\leftarrow$ Raw materials

Review: Coffee Shop Example

Matrix A (Semi-finished $\leftarrow$ Raw materials):

	 Milk	 Coffee	 Tea
$A =$  Leaf	0	0	2
 Lemon	0	0	1
 Bean	0	2	0
 Cow	1	0	0

Interpretation:

- To make  : need 1 
- To make  : need 2 
- To make  : need 2  + 1 

Review: Coffee Shop Example (continued)

Matrix B (Final products $\leftarrow$ Semi-finished):

		 Set 1	 Set 2
$B =$	 Milk	2	1
	 Coffee	0	2
	 Tea	1	1

Interpretation:

-  : $2 \times$  + $0 \times$  + $1 \times$ 
-  : $1 \times$  + $2 \times$  + $1 \times$ 

Review: Coffee Shop Example (result)

Matrix $C = AB$ (Final products $\leftarrow$ Raw materials):

		 Set 1	 Set 2
$C =$	 Leaf	2	2
	 Lemon	1	1
	 Bean	0	4
	 Cow	2	1

Interpretation:

-  : needs 2  + 1  + 0  + 2 
-  : needs 2  + 1  + 4  + 1 

Where Each Concept Appears

Concept	Matrix A	Matrix B	Matrix C
Raw materials 	rows	—	rows
Semi-finished 	columns	rows	—
Final products 	—	columns	columns

The Fundamental Principle

When we **redefine** one of these three concepts (change the unit, repackage, bundle), all matrices that reference this concept must update their representation. But the underlying production relationships remain unchanged.

Row Operations: Contravariant Logic

Row Operations: Redefining Raw Materials

Core Idea: Row operations correspond to **redefining raw materials**.

When we redefine a raw material (change its unit or bundle it with others), we must update:

- The **rows of matrix A** (since raw materials appear in rows of A)
- The **rows of matrix C** (since raw materials appear in rows of C)
- Matrix B is **not affected** (semi-finished products don't change)

The Subtle Point

Row operations have **contravariant (inverse) logic** because rows and materials have a **multiplication relationship**:

$$(\text{demand coefficient}) \times (\text{material}) = (\text{total amount needed})$$

Why Contravariant Logic?

The key equation:

$$\text{demand} \times \text{material} = \text{constant}$$

Example with  :

- Original: need 4  for 
- If we redefine:  $\rightarrow$   (double bean, one unit = two old beans)
- Then we need: 2   = 4 old  (same total!)

Contravariant = Inverse Direction

- Material becomes **larger** $\Rightarrow$ Need **fewer** units
- Material becomes **smaller** $\Rightarrow$ Need **more** units

Type 1: Row Multiplication

Definition (Row Multiplication)

Multiply a row by a nonzero number k .

Notation: $\text{row} \times k$

What it means: Change the unit size of a material.

Example:  $\rightarrow$  (double bean)

Contravariant Logic

- Material unit becomes **larger** ($\times 2$)
- Demand numbers become **smaller** ($\times \frac{1}{2}$)
- Row operation: multiply by $\frac{1}{2}$

Example 1.1: → (Double Bean)

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

What we're doing: Redefine  →  

One   = two old  . The new unit is **larger**!

Example 1.1: How Many Do We Need?

For 🍲 Set 2, originally:

$$2 \text{ 🌿} + 1 \text{ 🍋} + 4 \text{ 🍲} + 1 \text{ 🐮}$$

After redefining 🍲 $\rightarrow$ 🍲 🍲 :

- Still need: 2 🌿 , 1 🍋 , 1 🐮 (unchanged)
- For beans: originally 4 🍲
- How many 🍲 🍲 ? Since 1 🍲 🍲 = 2 🍲
- Need: $\frac{4 \text{ 🍲}}{2} = 2 \text{ 🍲 🍲}$

New recipe: 2 🌿 + 1 🍋 + 2 🍲 🍲 + 1 🐮

Number changed from 4 $\rightarrow$ 2!

Example 1.1: Result

Updated table (row 3 **divided by 2**):

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Double bean	0	2
 Cow	2	1

Matrix notation: Perform $r_3 \rightarrow \frac{1}{2}r_3$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{r_3 \rightarrow \frac{1}{2}r_3} \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 2 \\ 2 & 1 \end{pmatrix}$$

Key

Material $\times 2 \rightarrow$ Row $\times \frac{1}{2}$ (contravariant!)

Type 2: Row Switching

Definition (Row Switching)

Swap row i and row j .

Notation: $r_i \leftrightarrow r_j$

Interpretation: Reordering materials in the table.

This doesn't change any production relationships, just the order.

Example:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{r_1 \leftrightarrow r_2} \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

Type 3: Row Addition

Definition (Row Addition)

Add a multiple of one row to another row.

Notation: row $i + k \times$ row j

What it means: Bundle materials together.

Example: 🍋 $\rightarrow$ 🍋 + 2 🌿 (lemon bundled with 2 leaves)

Contravariant Logic - The Tricky Part!

- New 🍋 **includes** 2 🌿 (material change: +2 🌿)
- So we need **fewer** separate 🌿 (row change: -2 🍋)
- **Opposite signs!** Material +, Row -

Example 1.3: 🍋 → 🍋 + 2🌿

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

What we're doing: Bundle 🍋 with 2🌿

Now each 🍋 comes with 2🌿 included (like a combo pack!)

Example 1.3: How Many Do We Need?



For Set 1, originally:

$$2 \text{ 🌿} + 1 \text{ 🍊} + 0 \text{ 🍷} + 2 \text{ 🐮}$$

After bundling $\text{🍊} \rightarrow \text{🍊} + 2 \text{ 🌿}$:

- Still need: 1 🍊 (bundled), 0 🍷 , 2 🐮
- The 1 bundled 🍊 **includes** 2 🌿
- Originally needed: 2 🌿 total
- **Already have** 2 🌿 from the bundled 🍊 !
- Need **0 extra** 🌿

$$\text{New recipe: } 0 \text{ 🌿} + 1(\text{🍊} + 2 \text{ 🌿}) + 0 \text{ 🍷} + 2 \text{ 🐮}$$

Leaf number changed from 2 $\rightarrow$ 0!

Example 1.3: Result

Updated table (row operation: $r_1 \rightarrow r_1 - 2r_2$):

	 Set 1	 Set 2
 Leaf	0	0
 +2 	1	1
 Bean	0	4
 Cow	2	1

Key - Contravariant!

 includes 2  $\rightarrow$ need **fewer** separate  (subtract!)

Summary: Contravariant Logic for Rows

Material Change	Row Change
 →   (larger)	row $\times \frac{1}{2}$ (smaller)
 →  + 2  (includes )	 row - 2 $\times$  row

Remember

Rows work **backwards** from the material change!

Column Operations: Covariant Logic

Column Operations: Redefining Products

Core Idea: Column operations correspond to **redefining final products**.

When we redefine a final product (change its portion size or bundle it), we must update:

- The **columns of matrix B** (since products appear in columns of B)
- The **columns of matrix C** (since products appear in columns of C)
- Matrix A is **not affected** (raw materials don't change)

The Crucial Difference

Column operations have **covariant (direct) logic** because columns and products have an **equation relationship**:

$$(\text{product}) = (\text{recipe})$$

They appear on opposite sides of the equals sign!

Why Covariant Logic?

The key equation:

$$\text{product} = \text{recipe}$$

Example with :

- Original : needs $2 \text{ 🌿} + 1 \text{ 🍊} + 2 \text{ 🐮}$
- If we make: $2 \times$  (double portion)
- Then need: $2 \times (2 \text{ 🌿} + 1 \text{ 🍊} + 2 \text{ 🐮})$
- $= 4 \text{ 🌿} + 2 \text{ 🍊} + 4 \text{ 🐮}$ (**everything doubles!**)

Covariant = Same Direction

- Product becomes **larger** $\Rightarrow$ Recipe becomes **larger**
- Product becomes **smaller** $\Rightarrow$ Recipe becomes **smaller**

Type 1: Column Multiplication

Definition (Column Multiplication)

Multiply a column by a nonzero number k .

Notation: column $\times k$

What it means: Make a bigger or smaller portion.

Example:  $\rightarrow 2 \times$  (double portion)

Covariant Logic

- Product becomes **larger** ($\times 2$)
- Recipe numbers become **larger** ($\times 2$)
- Column operation: multiply by 2

Example 1.4:



Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

What we're doing: Make 2x  (double portion)

What happens?

- Original : 2  + 1  + 0  + 2 
- Double : 2x everything = 4  + 2  + 0  + 4 

Example 1.4: Result

Updated table (column 1 **multiplied by 2**):

	 	Double Set 1	 Set 2
 Leaf		4	2
 Lemon		2	1
 Bean		0	4
 Cow		4	1

Matrix notation: Perform $c_1 \rightarrow 2c_1$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_1 \rightarrow 2c_1} \begin{pmatrix} 4 & 2 \\ 2 & 1 \\ 0 & 4 \\ 4 & 1 \end{pmatrix}$$

Key

Product $\times 2 \rightarrow$ Column $\times 2$ (**same direction!**)

Type 2: Column Switching

Definition (Column Switching)

Swap column i and column j .

Notation: $c_i \leftrightarrow c_j$

Interpretation: Reordering products in the table.

Example:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_1 \leftrightarrow c_2} \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 4 & 0 \\ 1 & 2 \end{pmatrix}$$

Type 3: Column Addition

Definition (Column Addition)

Add a multiple of one column to another column.

Notation: column $j + k \times$ column i

What it means: Bundle products together.

Example:  $\rightarrow$  +  (combo meal!)

Covariant Logic - Straightforward!

- New  **includes**  (product change: + )
- Recipe needs ingredients for **both** (recipe change: +  recipe)
- **Same signs!** Product +, Recipe +

Example 1.6: → +

Original table:

	 Set 1	 Set 2
 Leaf	2	2
 Lemon	1	1
 Bean	0	4
 Cow	2	1

What we're doing: Bundle  +  (combo!)

Original recipes:

-  : 2  + 1  + 0  + 2 
-  : 2  + 1  + 4  + 1 

Example 1.6: How Much Do We Need?

For the combo  + :

-  needs: 2  + 1  + 4  + 1 
-  needs: 2  + 1  + 0  + 2 
- **Add them together:**
 - Leaf: $2 + 2 = 4$ 
 - Lemon: $1 + 1 = 2$ 
 - Bean: $4 + 0 = 4$ 
 - Cow: $1 + 2 = 3$ 

New recipe: 4  + 2  + 4  + 3 

All numbers increased! (covariant)

Example 1.6: Result

Updated table (column 2: $c_2 \rightarrow c_2 + c_1$, **add not subtract**):

	 Set 1	 Combo
 Leaf	2	4
 Lemon	1	2
 Bean	0	4
 Cow	2	3

Matrix notation: Perform $c_2 \rightarrow c_2 + c_1$:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix} \xrightarrow{c_2 \rightarrow c_2 + c_1} \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 0 & 4 \\ 2 & 3 \end{pmatrix}$$

Key

Summary: Covariant Logic for Columns

Product Change	Column Change
 $\rightarrow 2 \times$  (larger)	column $\times 2$ (larger)
 $\rightarrow$  $+$  (includes )	 col $+$  col

Remember

Columns work **forwards** with the product change!

THE KEY DISTINCTION

	Rows	Columns
Logic	Contravariant (inverse)	Covariant (direct)
Example	 → 	 → 2 × 
Change	material $\times 2$ row $\times \frac{1}{2}$	product $\times 2$ column $\times 2$

The Essence

Rows work backwards. Columns work forwards.

Why We Don't Update Semi-finished Products

Semi-finished products ( ,  , ) appear in:

- **Columns of A** (as outputs) → would need column operations (covariant)
- **Rows of B** (as inputs) → would need row operations (contravariant)

The Problem

Updating semi-finished products would require:

- **Two different types of operations**
- With **opposite logic** (contravariant vs covariant)
- On **two different matrices** (both A and B)

This is significantly more complex than row-only or column-only operations.

Solution: We avoid this complication and focus on raw materials (rows)

Simultaneous Operations

Preserve Equations

Change of Perspective

What we've learned so far:

- **Why** rows are contravariant (multiplication relationship)
- **Why** columns are covariant (equation relationship)
- **How** to update tables when redefining materials/products

From This Point Forward

We **forget about materials/products** and focus only on:

- **What:** Matrices A , B , C
- **How:** Simultaneous operations keep equations valid

This simplifies our work: we only track **matrix changes**, not material redefinitions.

Proposition 1.1: Row Operation Invariance

Proposition (Row Operation Invariance)

In the equation $C = AB$, if we perform the **same row operation** on both A and C simultaneously, the equality remains valid.

Why? Raw materials appear in the **rows** of both A and C . When we redefine raw materials, both must update. Matrix B doesn't reference raw materials, so it doesn't change.

But from now on, we don't need to think about "why" — we just use the fact that **simultaneous row operations on A and C preserve the equation**.

Example 1.7: Row Operation Preserves Equation

Consider:

$$\underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 2 & 2 \\ 2 & 3 \end{pmatrix}}_C$$

Perform $r_1 \rightarrow 2r_1$ on **both A and C**:

$$\underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}}_{A'} \underbrace{\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 4 & 4 \\ 2 & 3 \end{pmatrix}}_{C'}$$

Still valid! ✓

Note: B is unchanged because it doesn't have raw materials in its rows.

Proposition 1.2: Column Operation Invariance

Proposition (Column Operation Invariance)

In the equation $C = AB$, if we perform the **same column operation** on both B and C simultaneously, the equality remains valid.

Why? Final products appear in the **columns** of both B and C . When we redefine final products, both must update. Matrix A doesn't reference final products, so it doesn't change.

But from now on, we don't need to think about products — we just use the fact that **simultaneous column operations on B and C preserve the equation**.

Example 1.8: Column Operation Preserves Equation

Consider:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}}_B = \underbrace{\begin{pmatrix} 2 & 1 \\ 2 & 6 \end{pmatrix}}_C$$

Perform $c_1 \rightarrow 3c_1$ on **both B and C**:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} 6 & 1 \\ 3 & 3 \end{pmatrix}}_{B'} = \underbrace{\begin{pmatrix} 6 & 1 \\ 6 & 6 \end{pmatrix}}_{C'}$$

Still valid! ✓

Note: A is unchanged because it doesn't have final products in its columns.

Why These Propositions Matter

Foundation for Matrix Equations

Propositions 1.1 and 1.2 provide the theoretical basis for:

- Solving matrix equations $AB = C$
- Finding matrix inverses A^{-1}
- Understanding how operations affect equations

Key insight: We can **manipulate** equations $C = AB$ using row/column operations, just like we manipulate algebraic equations, as long as we:

- Apply row operations to **both A and C**
- Apply column operations to **both B and C**

Matrix Inverses

Definition: Matrix Inverse

Definition (Matrix Inverse)

An $n \times n$ matrix A is called **invertible** if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the $n \times n$ identity matrix. We call B the **inverse** of A and denote it A^{-1} .

Interpretation: If A is the recipe "raw materials $\rightarrow$ products", then A^{-1} is the recipe "products $\rightarrow$ raw materials".

Example 1.9: Coffee Shop Inverse

Consider:

$$A = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = \begin{array}{c|cc} & \img alt="coffee cup icon" data-bbox="540 185 585 240"/> \text{Coffee} & \img alt="milk glass icon" data-bbox="700 185 735 240"/> \text{Milk} \\ \hline \img alt="bean icon" data-bbox="395 260 440 315"/> \text{Bean} & 0 & 2 \\ \img alt="cow icon" data-bbox="395 335 445 400"/> \text{Cow} & 1 & 0 \end{array}$$

Then:

$$A^{-1} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{array}{c|cc} & \img alt="bean icon" data-bbox="560 515 605 575"/> \text{Bean} & \img alt="cow icon" data-bbox="705 510 755 575"/> \text{Cow} \\ \hline \img alt="coffee cup icon" data-bbox="400 590 445 645"/> \text{Coffee} & 0 & 1 \\ \img alt="milk glass icon" data-bbox="420 665 450 730"/> \text{Milk} & \frac{1}{2} & 0 \end{array}$$

Verification:

$$AA^{-1} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Proposition 1.3: Inverse is Unique

Proposition (Uniqueness of Inverse)

If A has an inverse, it is unique.

Proof: Suppose B and C both satisfy the definition. Then:

$$B = BI = B(AC) = (BA)C = IC = C$$

Therefore $B = C$. □

Proposition 1.4: Product of Invertible Matrices

Proposition (Inverse of Product)

If A and B are invertible, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

Verification:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Order Reversal

$(AB)^{-1} = B^{-1}A^{-1}$, **not** $A^{-1}B^{-1}$.

Intuition: Socks then shoes $\rightarrow$ undo by removing shoes first, then socks.

Why Introduce Inverses?

Why Do We Need Matrix Inverses?

Question: Why introduce the concept of inverse matrices?

Answer: Inverses provide **notational convenience** for tracking invariants when solving matrix equations.

The Key Insight

Writing expressions like $B = A^{-1}C$ or $B = CA^{-1}$ makes explicit that:

- " **B doesn't change**" when we perform operations
- We can **simultaneously transform** the " A^{-1} " part and the " C " part
- The transformation preserves the equation

The next two theorems are the **most important** results for practical computation!

Theorem 1.6: Row Operations on $A^{-1}C$

Proposition (Row Operations Preserve $A^{-1}C$)

In the expression $B = A^{-1}C$, if we perform the **same row operation** on both "A" and "C" simultaneously, we get:

$$B = (A')^{-1}C'$$

where A' , C' are the matrices after the row operation, and **B remains unchanged.**

Why? From $AB = C$, applying row operations to both A and C preserves the equation (Prop 1.1):

$$A'B = C'$$

Therefore: $B = (A')^{-1}C'$

Key Application

This theorem allows us to solve $AB = C$ for B by performing row operations!

Example 1.10: Using $A^{-1}C$ Row Operations (Setup)

Problem: Solve $AB = C$ for B where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Method: Write $B = A^{-1}C$ and perform row operations to transform " A " into I :

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Goal: Transform " A " to " I " using row operations.

Example 1.10: Step 1-2

Step 1: $r_1 \leftrightarrow r_2$

$$B = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ 5 & 8 \end{pmatrix}$$

Step 2: $r_2 \rightarrow r_2 - 2r_1$

$$B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ -1 & -2 \end{pmatrix}$$

Example 1.10: Step 3-4

Step 3: $r_2 \rightarrow (-1) \cdot r_2$

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 & 5 \\ 1 & 2 \end{pmatrix}$$

Step 4: $r_1 \rightarrow r_1 - r_2$

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = I^{-1} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Since $I^{-1} = I$:

$$B = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Example 1.10: Verification

Check: Does $AB = C$?

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4+1 & 6+2 \\ 2+1 & 3+2 \end{pmatrix} = \begin{pmatrix} 5 & 8 \\ 3 & 5 \end{pmatrix}$$

Yes! ✓

Key Observation

We transformed "A" to "I" using row operations, and automatically obtained B!

Theorem 1.7: Column Operations on CB^{-1}

Proposition (Column Operations Preserve CB^{-1})

In the expression $A = CB^{-1}$, if we perform the **same column operation** on both "C" and "B" simultaneously, we get:

$$A = C'(B')^{-1}$$

where C' , B' are the matrices after the column operation, and **A remains unchanged**.

Why? From $AB = C$, applying column operations to both B and C preserves the equation (Prop 1.2):

$$AB' = C'$$

Therefore: $A = C'(B')^{-1}$

Key Application

This theorem allows us to solve $AB = C$ for A by performing column operations!

Example 1.11: Using CB^{-1} Column Operations (Setup)

Problem: Solve $AB = C$ for A where

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 5 & 3 \\ 8 & 5 \end{pmatrix}$$

Method: Write $A = CB^{-1}$ and perform column operations to transform " B " into I :

$$A = \begin{pmatrix} 5 & 3 \\ 8 & 5 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1}$$

Goal: Transform " B " to " I " using column operations.

Example 1.11: Step 1-2

Step 1: $c_1 \leftrightarrow c_2$

$$A = \begin{pmatrix} 3 & 5 \\ 5 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-1}$$

Step 2: $c_2 \rightarrow c_2 - 2c_1$

$$A = \begin{pmatrix} 3 & -1 \\ 5 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}^{-1}$$

Example 1.11: Step 3-4

Step 3: $c_2 \rightarrow (-1) \cdot c_2$

$$A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1}$$

Step 4: $c_1 \rightarrow c_1 - c_2$

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} I^{-1}$$

Wait, error! Let me recalculate...

Actually:

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$$

Summary: The Two Critical Theorems

Theorem 1.6: Row Operations on $A^{-1}C$

Simultaneous row operations on "A" and "C" in $B = A^{-1}C$ preserve B.

Use case: Solving $AB = C$ for B

Theorem 1.7: Column Operations on CB^{-1}

Simultaneous column operations on "C" and "B" in $A = CB^{-1}$ preserve A.

Use case: Solving $AB = C$ for A

Why These Matter

These two theorems are the **foundation** for:

- Finding matrix inverses
- Solving matrix equations
- Understanding matrix transformations

Finding Matrix Inverses

Finding A^{-1} : The Method

Goal: Find A^{-1} for a given matrix A .

Idea: Solving $AA^{-1} = I$ for A^{-1} is a special case of solving $AB = C$.

From $A^{-1} = A^{-1}I$, perform **row operations** to transform " A " into I :

$$A^{-1} = A^{-1}I \xrightarrow{\text{row ops}} I^{-1} \cdot (\text{result}) = \text{result}$$

Algorithm:

1. Write $A^{-1} = A^{-1}I$
2. Apply row operations to both " A " and " I " until " A " becomes I
3. The result on the right is A^{-1}

Example 1.12: Finding Inverse (Row Operations)

Find: A^{-1} where $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$

Setup:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 1: $r_2 \rightarrow r_2 - r_1$

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

Example 1.12: Steps 2-5

Step 2: $r_1 \leftrightarrow r_2$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix}$$

Step 3: $r_2 \rightarrow r_2 - 2r_1$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ 3 & -2 \end{pmatrix}$$

Step 4: $r_2 \rightarrow (-1) \cdot r_2$

$$A^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} -1 & 1 \\ -3 & 2 \end{pmatrix}$$

Step 5: $r_1 \rightarrow r_1 - r_2$

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

Example 1.13: Finding Inverse (Column Operations)

Alternative: Use column operations on $IA^{-1} = A^{-1}$

Setup:

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}^{-1}$$

Step 1: $c_1 \rightarrow c_1 - c_2$

$$A^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}^{-1}$$

Step 2: $c_2 \rightarrow c_2 - c_1$

$$A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^{-1}$$

Example 1.13: Final Step

Step 3: $c_1 \rightarrow c_1 - c_2$

$$A^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$$

Same answer! ✓

Observation

Both row operations and column operations give the same inverse.

When Does an Inverse Exist?

Not All Matrices Are Invertible

Example:

$$A = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$$

Observation: Row 2 = 2 × Row 1 (redundant information)

Try to find inverse:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Perform $r_2 \rightarrow r_2 - 2r_1$:

$$A^{-1} = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

Stuck!

Cannot transform "A" to I. This matrix has **no inverse**.

Invertibility Criterion (Preview)

Criterion

An $n \times n$ matrix is invertible if and only if it can be transformed to I using row operations.

Equivalently: **rank** = n .

Detailed discussion of rank and solvability conditions will be covered in **Chapter 2**.

For now:

- If row operations can reduce A to I : invertible
- If we get a zero row: not invertible

Visual Summary: Invertible vs Non-invertible

Invertible	Non-invertible
$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$	$\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$
Can reduce to I	Gets zero row
Rows are independent	Rows are dependent
Full rank	Not full rank
Unique inverse A^{-1}	No inverse

Exercises

Exercise 1.1

Consider the equation $AB = C$ where

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 4 & 7 \\ 3 & 5 \\ 2 & 3 \end{pmatrix}$$

Solve for B using row operations on $B = A^{-1}C$.

Exercise 1.2

Find the inverses using row operations:

$$(a) A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$$

$$(b) A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Exercise 1.3

Find the inverse of $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$ using **column operations**.

Verify it matches the result from row operations.

Exercise 1.4

Try to find the inverse of $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

What happens? Explain why this matrix is not invertible.

Exercise 1.5

Show that if A and B are invertible, then:

(a) $(A^{-1})^{-1} = A$

(b) $(A^T)^{-1} = (A^{-1})^T$

Exercise 1.6

Consider the coffee shop example with matrix

$$C = \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & 4 \\ 2 & 1 \end{pmatrix}$$

- (a) Perform row operation $r_1 \rightarrow r_1 - 2r_2$ and interpret the material redefinition.
- (b) Perform column operation $c_2 \rightarrow c_2 + c_1$ and interpret the product redefinition.

Summary: Key Takeaways

1. **Row operations: Contravariant** (inverse logic)
 - Material $\times k \rightarrow$ Row $\times \frac{1}{k}$ (reciprocal)
 - Material $x_i \rightarrow x_i + kx_j \rightarrow$ Row $r_j \rightarrow r_j - kr_i$ (opposite sign!)
2. **Column operations: Covariant** (direct logic)
 - Product $\times k \rightarrow$ Column $\times k$ (same)
 - Product $y_j \rightarrow y_j + ky_i \rightarrow$ Column $c_j \rightarrow c_j + kc_i$ (same sign!)
3. **Equation $C = AB$ is invariant** under:
 - Simultaneous row ops on A and C (Prop 1.1)
 - Simultaneous column ops on B and C (Prop 1.2)

Summary: Critical Theorems

4. **Left inverse = right inverse** for square matrices (Prop 1.4)
5. **Row operations on $A^{-1}C$ preserve B** (Theorem 1.6)
 - Use to solve $AB = C$ for B
6. **Column operations on CB^{-1} preserve A** (Theorem 1.7)
 - Use to solve $AB = C$ for A
7. **Finding A^{-1}** : use row ops on $A^{-1}I$ or column ops on IA^{-1} until " A " becomes I

Remember

Rows work backwards. Columns work forwards.